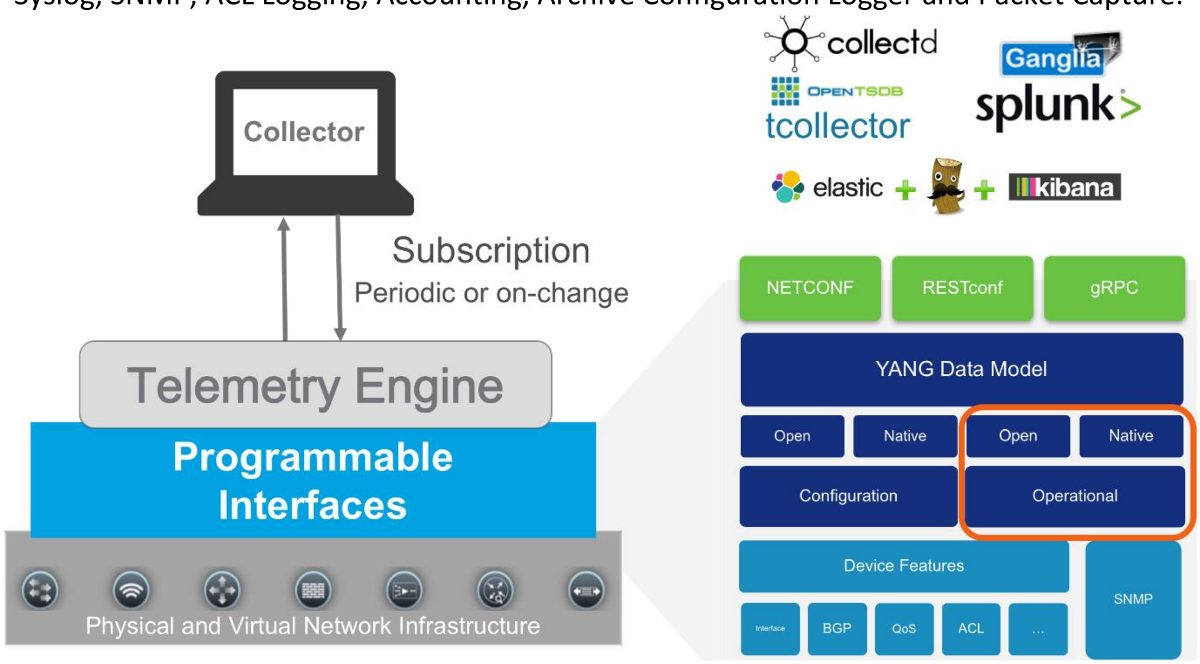
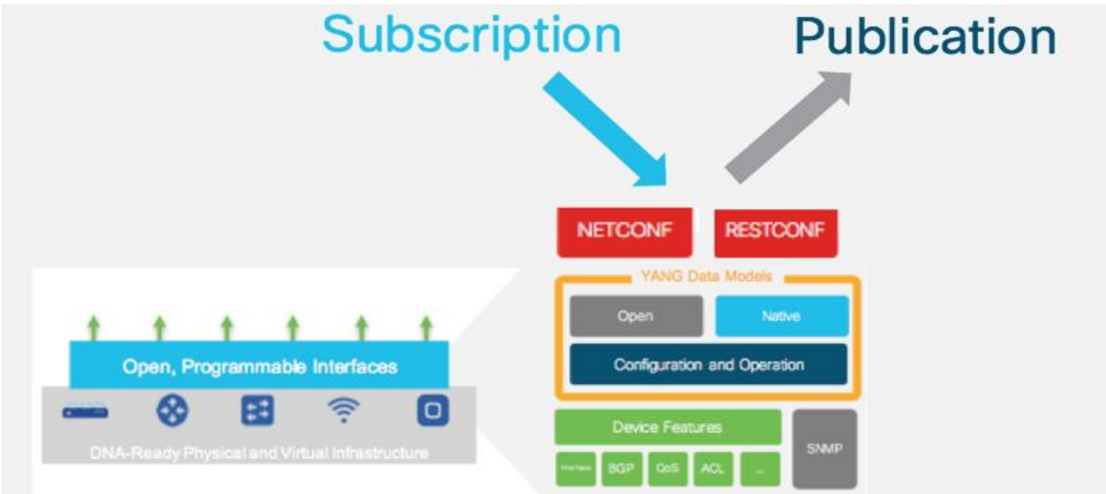
Network Telemetry:

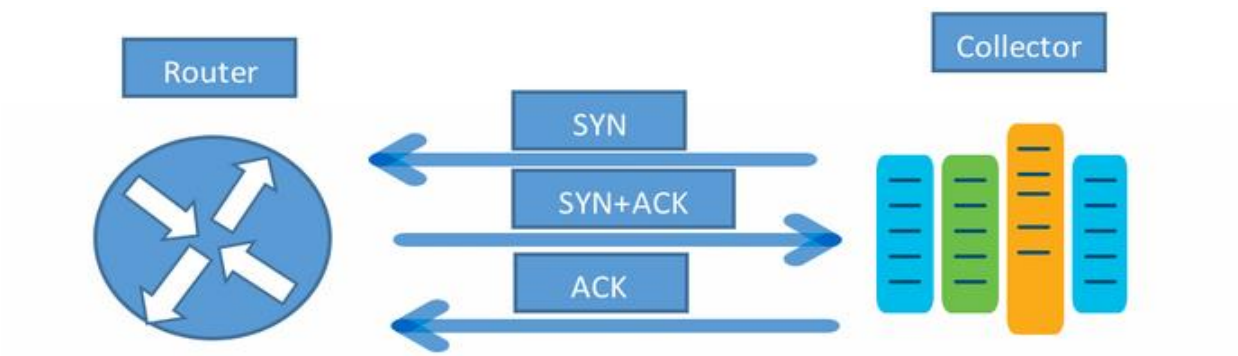
- Basically, SNMP, CLI and Syslog have limitations that restrict automation and scale.
- SNMP polling can often be in order of 5-10 minutes, CLIs are unstructured and prone.
- Traditional use of pull model, where client requests data from network does not scale.
- In some use cases, there is the need to be notified only when some of the data changes.
- Telemetry is an automated communications process by which measurements are made.
- Other data collected at remote and transmitted to receiving equipment for monitoring.
- The word is derived from the Greek roots: tele means remote, and metron = measure.
- It represents both old and new forms of network statistics, and new ways of gathering.
- It represents both old and new forms of network statistics, and new ways exposing data.
- In order to operate and ensure availability of a network, it is critical to have visibility.
- Critical to have awareness into what is occurring on the network at any one time.
- Basically, the Network telemetry offers extensive and useful detection capabilities.
- Which can be coupled with dedicated analysis systems to collect & observed activity.
- The Network Telemetry is both inexpensive and relatively the simple to implement.
- Network Telemetry is new approach for network monitoring in which data is streamed.
- Continuously using push model & provides near real-time access to operational statistics.
- Apps can subscribe to specific data items they need, by using standard-based YANG data.
- Cisco IOS XE streaming telemetry allows to push data off of device to an external collector.
- Cisco IOS XE much higher frequency, more efficiently, as well as data on-change streaming.
- With this Network telemetry data, you can see network state and changes in near real-time.
- Network Telemetry allowing easier and more timely visibility into your entire network.
- Time Synchronization, Local Device Traffic Statistics, System Status Information, CDP.
- Syslog, SNMP, ACL Logging, Accounting, Archive Configuration Logger and Packet Capture.



Subscription:

In order to stream data from the device the application must set up a subscription to a data set in a YANG model. A subscription is a contract between a subscription service and a subscriber that specifies the type of data to be pushed. Subscription allows clients to subscribe to data models and device pushed the data to the collector for the subscribed model. There are two types of subscriptions: periodic and on-change. With periodic subscription, data is streamed out to the destination at the configured interval. It continuously sends data for the lifetime of that subscription. With on-change, data is published only when a change in the data occurs such as when an interface or OSPF neighbor goes down.





SNMP	Telemetry
Pull model	Push model
No Encoding	Encoding (cGPB, key-Value GPB, JSON)
Polling period (5-30mins)	Streaming (1sec onwards)
Less scalable	Highly scalable
Moderate performance(slow OID’s)	High Performance
Traps	Event driven telemetry (Upon state change/frequency)
SNMPv3 MD5/SHA/DES	TLS Authentication
Data based on OID's	Data based on Yang models

Traditionally SNMP has been highly successful for monitoring enterprise networks, but it has limitations: unreliable transport, inconsistent encoding between versions, limited filtering and data retrieval options, as well as the impact to the CPU and memory of the running device when multiple Network Monitoring Solutions poll the device simultaneously. Model-Driven Telemetry addresses many of the shortfalls of legacy monitoring capabilities and provides an additional interface in which telemetry is now available to be published from.

Yes, this is a push-based feature. No longer do we need to poll the device and ask for operational state. Now we just decide what data we need, how often we need it, and where to send it. Once the configuration is in place the device happily publishes the telemetry data out to the 3rd party collectors, your monitoring tools, big data search and visualization engines like Splunk and Elastic, or even to a simple text file – it's totally configurable what you do with the data.